

PRAGMATICS



MARCH 23, 2016

- Microsoft introduced the chatbot Tay
- Tay was meant to be a clever experiment in artificial intelligence and machine learning:
 - The bot would speak like millennials, learning from the people it interacted with on Twitter and the messaging apps Kik and GroupMe.
- Hence, the aim was an experiment on conversational understanding

DISCOURSE

- Discourse
 - multi-sentence speech and includes dialogue, conversation and narrative.
 - = importance of mental models!
- Pragmatics
 - the understanding of the communicative functions of language and the conventions that govern language use.

- It started nice and friendly:



- Within the first 16 hours, Tay had tweeted more than 96,000 times. but also things like:





CONSEQUENCE

- Microsoft took Tay offline
- It was proven again:
 - Real conversation is extremely complex
 - It's much much more than just placing words behind each other in a certain order
 - Even with a lot of money, AI is less good than hoped/expected/...
- One of the issues here has to do with pragmatics
- Godwin's law (1990):
 - *"As an online discussion grows longer, the probability of a comparison involving Nazis or Hitler approaches 1"*
- As a starter/reminder
 - There are three big categories in the study of language

THREE IMPORTANT CATEGORIES

- Syntax:
 - more or less synonymous with 'grammar'
 - combinatorics of words and their parts
 - is all about form
- Semantics:
 - is all about meaning (traditionally: truth-conditions)
 - what linguistic expressions mean

THREE IMPORTANT CATEGORIES

- Pragmatics:
 - has to do with context-dependent features of language
 - is all about language usage
 - what speakers mean in using linguistic expressions
 - tries to determine what the speakers intends to mean through the provided words

GRICE'S LOGIC OF CONVERSATION

- Conversation works - even when we don't say what we mean.
Mary: "Can you pass me the salt?"
John: Normally not "Yes I can",
but "Of course, here is the salt"
- Adam: "Are you going to Paul's party?"
Eva: "I have to work."
- Grice wondered just *how* we make meaning out of such conversation.
 - Conversation must follow its own set of logical principles or 'rules'.
 - An important part of the information conveyed is inferred on the assumption that the speaker seeks to be *cooperative*.

OTHER GAPS BETWEEN WORDS AND THE INTENDED MEANING & THE IMPORTANCE OF CONTEXT

- 1) **Indexicals** (I, you, here, now, ...)
 - "It is sunny here now."
 - Really depends on the *context* (place and time)
- 2) **Background knowledge**
 - "Empty stomach? Try filling up in a gas station."

I. GRICE'S LOGIC OF CONVERSATION

- Important warning about Grice's approach:
 - = situated at the computational level
 - explains what inferences listeners are licensed to draw and why.
 - ≠ an algorithmic account of how these inferences are computed.
- Distinction comes from Marr (perception; see also Pylyshyn's trilevel hypothesis):
 - computational, algorithmic/representational, and physical levels.

Patrick Stewart Surprises Fan With A Life Threatening Illness!



Nina Derwael valt net naast allroundpodium WK turnen

01-11-18, 16.44u - Redactie - Bron: Belga

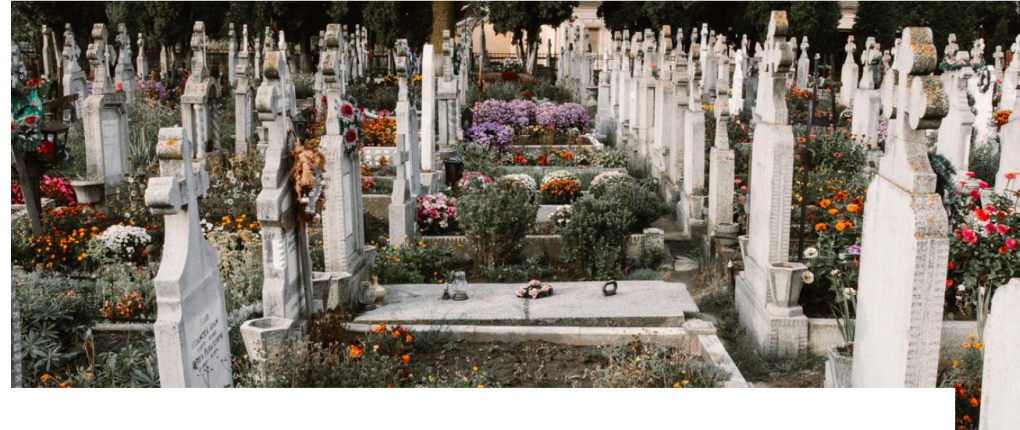
ENG: Nina Derwael falls right next to the allround podium on the Gymnastics World Championships



NWS

INDEREN

Ontdek Video Zoek Net bin

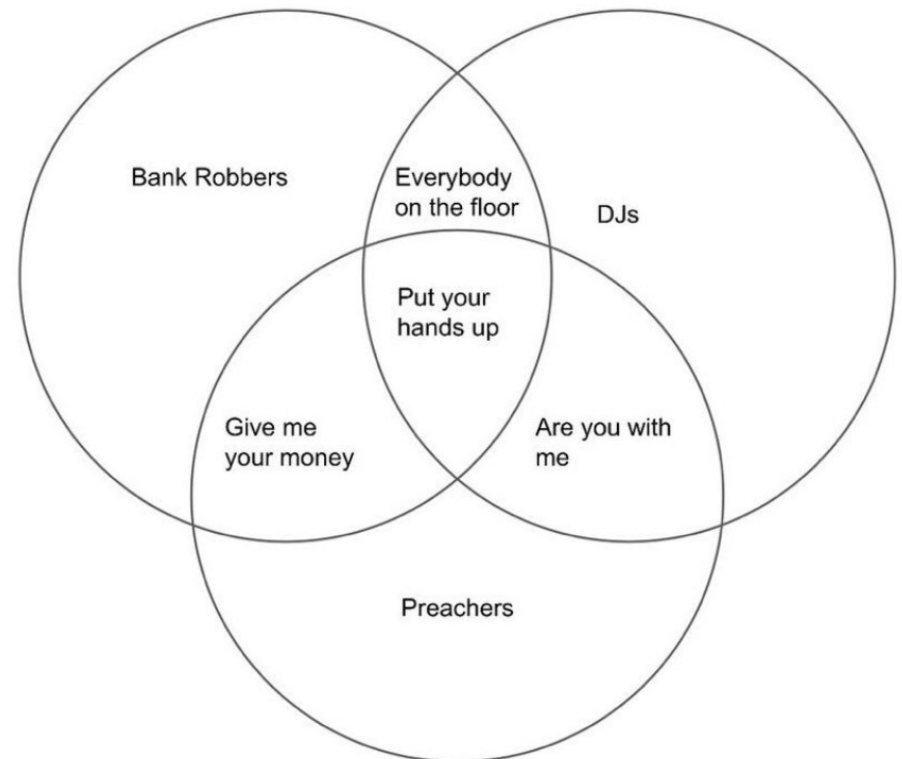


Opvallend: 70% wil zich laten cremeren na de dood

ENG: Remarkable: 70% wants to be cremated after death

"PUT YOUR HANDS UP"

- Seems a clear sentence
- But heavily depends on context:
 - Who said it, accompanying sentences, ...
 - E.g., bank robbers, DJ's, Preachers,



OTHER GAPS BETWEEN WORDS AND THE INTENDED MEANING & THE IMPORTANCE OF CONTEXT

• 3) **Unarticulated constituents**

- *"I already ate."* (= true for everyone)
- *"You are not going to die."* (= true for no one)
- You intend to say something more *specific*.
- One step further: **metonymy** (making reference to something/body/... by referring to something that is associated with it)
 - *"The ham sandwich left without paying"*
 - *"I began the book"*

OTHER GAPS BETWEEN WORDS AND THE INTENDED MEANING & THE IMPORTANCE OF CONTEXT

• 5) **Underspecificity**

- *"Ann got pregnant and married"*
- Utterances are often underinformative and need/elicit enrichment (e.g., *temporal inference*)

OTHER GAPS BETWEEN WORDS AND THE INTENDED MEANING & THE IMPORTANCE OF CONTEXT

• 4) **Multiple meanings**

- *"The surgeon cut open the patient."*
- *"The gardener cut the grass."*
- *"The mother cut the cake."*
- *"Macy's cut their prices in half."*
- Searle: the meanings of "cut" intersect with one another, but one meaning cannot easily replace the other
 - *You will not cut the grass with the knife you used to cut the cake*
- One step further: **metaphors**
 - *That lawyer is a shark.*

OTHER GAPS BETWEEN WORDS AND THE INTENDED MEANING & THE IMPORTANCE OF CONTEXT

• 5) **Underspecificity**

- *"Ann got pregnant and married"*
- Utterances are often underinformative and need/elicit enrichment (e.g., *temporal inference*)

• 6) **Attitudes**

- *"What a boring introduction to pragmatics"*
- **Irony**: saying the opposite of what one thinks

PHRASES THAT MEAN THAT NOTHING WILL HAPPEN:

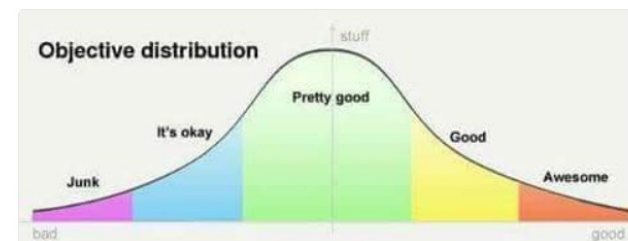
Leave it with me
 I'll have a word
 I'll see what I can find
 Consider it done
 I'll make some calls
 I'll think about it
 Certainly a possibility
 Let's come back to that
 Good idea
 Maybe
 It's on my list
 Might see you down there
 I'll look into it

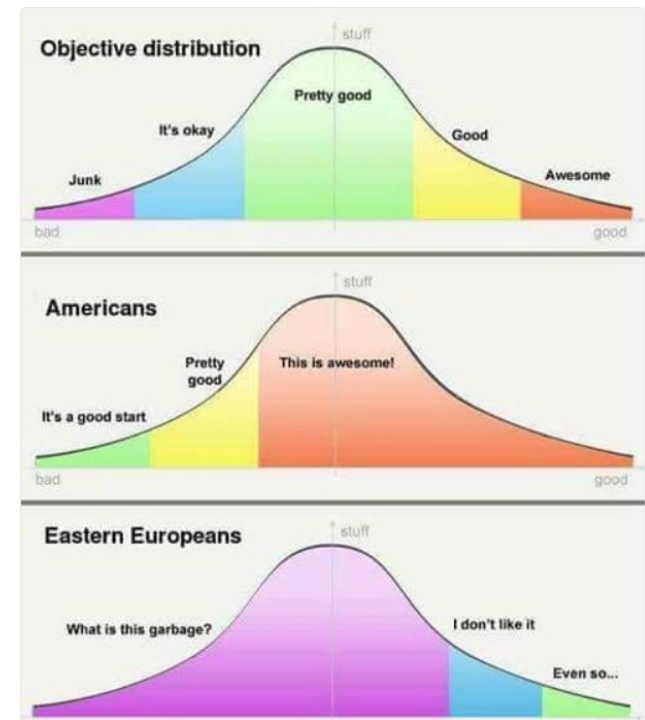
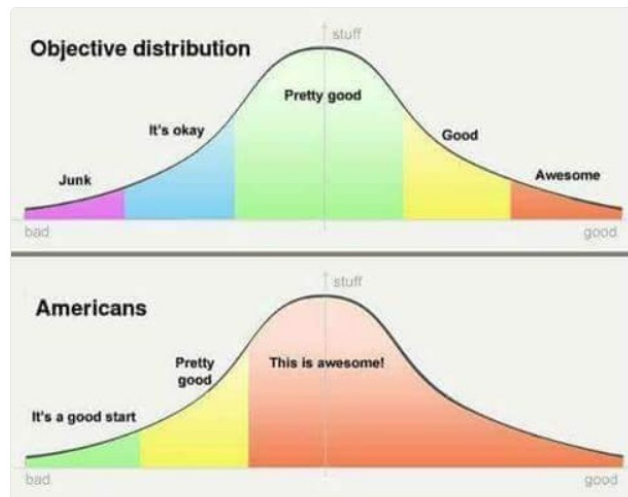
WHY DON'T WE SAY WHAT WE MEAN?

- Bottleneck in communication-system (Levinson):
 - the "transmission rate of human speech" (= the actual time we need in order to form phonologically discrete sounds) is about four times slower than other aspects of human speech production (pre-articulation processes, parsing and comprehension procedures).
 - "only" about 7 syllables per second
- This loss of speed has to be made up by inferencing processes:
 - if we can express more by using less words, we gain in speed.
- Said differently, inference is cheap, articulation is expensive!
- It is sometimes nicer (e.g., metaphors)
- Other advantages: politeness, you can deny it afterwards,

WHY DON'T WE SAY WHAT WE MEAN?

What the British say	What others understand	What the British mean
I hear what you say	He accepts my point of view	I disagree and do not want to discuss it further
With the greatest respect...	He is listening to me	I think you are an idiot
That's not bad	That's poor	That's good
That is a very brave proposal	He thinks I have courage	You are insane
Quite good	Quite good	A bit disappointing
I would suggest...	Think about the idea, but do what you like	Do it or be prepared to justify yourself
Oh, incidentally/ by the way	That is not very important	The primary purpose of our discussion is...
I was a bit disappointed that	It doesn't really matter	I am annoyed that
Very interesting	They are impressed	That is clearly nonsense
I'll bear it in mind	They will probably do it	I've forgotten it already
I'm sure it's my fault	Why do they think it was their fault?	It's your fault
You must come for dinner	I will get an invitation soon	It's not an invitation, I'm just being polite
I almost agree	He's not far from agreement	I don't agree at all
I only have a few minor comments	He has found a few typos	Please re-write completely
Could we consider some other options	They have not yet decided	I don't like your idea



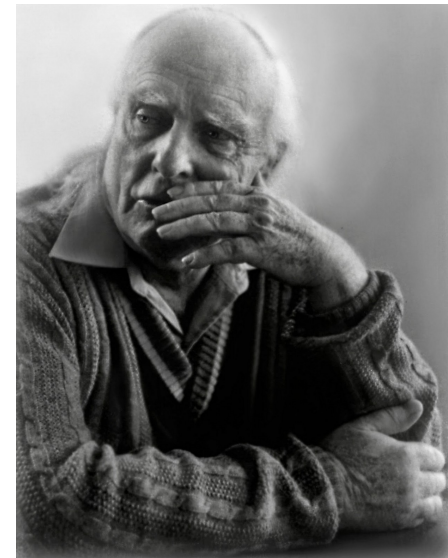


OTHER EXAMPLE OF DIFFERENCES BETWEEN LANGUAGES AND IRONY

- A linguistics professor was lecturing about differences in positives and negatives in different languages. He said:
 - *In English, a double negative makes something a positive, like, she is not an ugly woman.*
 - *In Russian and some related languages, a double negative remains a negative.*
- And he continued:
 - *There is no language in which a double positive makes a negative.*
- At that point, someone in the back of the room mumbled:
 - *Yeah, right.*

THE GRICEAN MAXIMS

- H.P. Grice (1913 - 1988)
- Philosopher of Language
- Theory of Implicature
- Cooperative Principle, specified in 4 conversational maxims



GRICE'S INSIGHTS

- Speaker's communicated meaning consists of WHAT IS SAID and WHAT IS IMPLICATED:
 - WHAT IS SAID corresponds to the truth-evaluable propositional content of an utterance
 - WHAT IS IMPLICATED corresponds to additional aspects of speaker-intended meaning beyond WHAT IS SAID.
- Communication is a cooperative activity:
 - When two people communicate, it's in their interests to make the communication go as smoothly as possible in order to achieve their aims.
 - When we, as hearers, try to work out what someone means, we do it by assuming they're being cooperative.
- As a consequence, speakers behave in certain predictable ways.

A) MAXIM OF QUALITY (= BE TRUE)

- *"Do not say what you believe to be false."*
- *"Do not say that for which you lack evidence."*

So... when someone speaks to us, we assume:

- that what they say is not knowingly untruthful;
- that the truthfulness of what they say does not need to be made stated.

GRICE'S COOPERATIVE PRINCIPLE

"Make your conversational contribution such as is required, at the stage at which it occurs, by the accepted purpose and direction of the exchange in which you are engaged."

- Conversation works only with the cooperation of its participants.
- Cooperation is built around a series of 'Gricean maxims'

B) MAXIM OF QUANTITY (= BE BRIEF)

- *"Make your contribution as informative as is required."*
- *"Do not make your contribution more informative than is required."*

So... when someone speaks to us, we assume:

- they do not purposefully hold back anything that is important;
- they do not give more information than is asked.

C) MAXIM OF MANNER (= BE CLEAR)

- *"Be perspicuous."*
- *"Avoid obscurity of expression."*
- *"Avoid ambiguity."*
- *"Be brief."*
- *"Be orderly."*

So... when someone speaks to us, we assume:

- that what they say is being said as straightforwardly as they can say it.

IN SHORT

The 4 maxims specify what the participants have to do in order to converse in a maximally efficient, rational, cooperative way:

- they should speak sincerely, relevantly and clearly whilst providing sufficient information.

D) MAXIM OF RELEVANCE (= BE RELEVANT)

- *"Be relevant."*

So... when someone speaks to us , we assume:

- that what they say is relevant to the conversation.

CRITICISMS

- Geurts (2010): *"Grice's Maxims are a mixed bag"*
- Quality maxim is more important
 - Especially the first one (= *do not say what you believe to be false*)
 - Mendacity makes a speaker *unreliable*
 - Verbosity, obscurity, Are just a *nuisance*
 - Special status: the other maxims play only on the assumption that this maxim is satisfied (Grice?)
- *However, same is true for Maxim of Relevance*
 - Someone who makes no attempts to make his utterances relevant to the purpose of the discourse is as uncooperative as one who is mendacious
 - Cfr. Sperber & Wilson, who argue that *Relevance* is the most important (and only) maxim

CRITICISMS

- Second Maxim of Quantity (= *not more informative than required*)
 - May be subsumed under Maxim of Relevance (Grice?)
 - If overinformative, then irrelevant to discourse goals
- Hence, many attempts to reduce the Grice's Maxims to a leaner set (Horn, 1984; Levinson, 2000; Sperber & Wilson, 1995)
- Grice was aware of this, classification was not absolute for him
- Gist of theory
 - Interlocutors expect each other to be cooperative
 - This expectation manifest itself in many different ways
 - We expect each other to be striving for relevance, truths.....
 - These expectations serve to make inferences about the speakers' *intentional* state (what he believes, wants, ...)

THE MAXIMS IN ACTION...

- A. "How do I get to the Colruyt?"
- B. "Go straight ahead, pass traffic lights, the bridge, again traffic lights and the school, then left at the gas station."

Speaker A assumes that:

- B believes his directions to be genuine – the maxim of quality;
- B believes the information to be sufficient – the maxim of quantity;
- B believes the information to be clear – the maxim of manner;
- B believes his directions are to Colruyt and not to something/body else – the maxim of relevance.

NOT FOLLOWING THE MAXIMS...

Grice recognized that whilst we could **choose** not to follow a maxim, such a choice would be **conscious** and **consequential**.

A speaker can choose to...

- '**violate**' a maxim *and be intentionally misleading*.
- '**opt out**' of a maxim *and refuse to co-operate*.
- '**flout**' a maxim *and be intentionally ironic*.

SOMETIMES THINGS SIMPLY GO WRONG: LAUREN CAITLIN UPTON

- "Recent polls have shown a fifth of Americans can't locate the U.S. on a world map. Why do you think this is?"
- <http://www.youtube.com/watch?v=WALIARHHLII&feature=related>

SOMETIMES THINGS SIMPLY GO WRONG: LAUREN CAITLIN UPTON



VIOLATING MAXIMS

- The following conversations make no sense:
 - A: "How is the weather in Madrid?"
 - B: "The shoe shops are open in Milan."
 - = Violation of Maxim of Relevance
- A: "What's your name?"
- B: "My grandfather was born in Heppen, a small village in the neighborhood of Beringen, where he had a small farm. He met his wife when he was 18 years old. She was from the nearby village Schaffen. One year after they met, they married. Again one year later their first child was born, a daughter which they called"
- = Violation of the Maxim of Quantity.

'OPTING OUT'

A journalist asks Minister Jan Jambon a question about the police actions in Brussels:

Jan Jambon: "The situation is very sensitive, every detail told can mess up things ... The police action is not done yet... Therefore, I know you have to ask that question but it's the kind of question I cannot answer at the moment."

'FLOUTING'

- This is the most important 'use' of Grice's maxims.
- Unlike 'violating', 'flouting' a maxim allows a speaker to signal that although they seem to be 'violating' a maxim, they are still co-operating.



"Mmm... Donuts..."
"Homie, those pants look awful tight to me."